WORKED SOLUTION

STEP II 2013, Question 11

Mechanics • Collision

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Read the question before you go on. It is in the STEP Support Programme's question database:

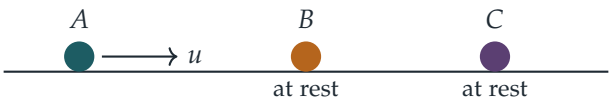
step.maths.org/questions/13-s2-q11

The key idea

Every collision here is between two equal masses, so each one is the same little calculation: conservation of momentum plus Newton's law of restitution. Do it once in general, then reuse it. Whether a further collision happens is just a question of whether the particle behind is moving faster than the particle in front, so the whole question comes down to comparing velocities.

Solution

Model the particles as identical particles of mass m on a smooth surface, so no horizontal forces act between collisions and each particle moves at constant velocity. Call them A, B, C in order along the line, with A the one projected towards B and C. Take the direction of A's initial motion as positive.



A general collision

Suppose a particle with velocity p catches up with an identical particle in front of it with velocity $q < p$, and afterwards they have velocities p' and q'. Momentum (cancelling m) and restitution give

$$p' + q' = p + q, \qquad q' - p' = e(p - q).$$

Adding and subtracting,

$$p' = \tfrac{1}{2}[(p + q) - e(p - q)], \qquad q' = \tfrac{1}{2}[(p + q) + e(p - q)]. \tag{$*$}$$

Since $e > 0$ and $p > q$, we always get $q' > p'$: after a collision the front particle moves faster than the rear one, so those two cannot meet again unless something else changes

one of their velocities.

Part (i)

First collision (A hits B): put $p = u$, $q = 0$ in $(*)$:

$$v_A = \tfrac{1}{2}u(1-e), \qquad v_B = \tfrac{1}{2}u(1+e).$$

Now $v_B > v_A$, so A cannot catch B; but B moves towards the stationary C, so the second collision is B with C.

Second collision (B hits C): put $p = \frac{1}{2}u(1+e)$, $q = 0$:

$$v_B = \tfrac{1}{2}\cdot\tfrac{1}{2}u(1+e)(1-e) = \tfrac{1}{4}u(1-e^2), \qquad v_C = \tfrac{1}{4}u(1+e)^2.$$

A is untouched, so it still has $v_A = \frac{1}{2}u(1-e)$. All three velocities are positive (as $0 < e < 1$), so these are the speeds $\frac{1}{2}u(1-e)$, $\frac{1}{4}u(1-e^2)$, $\frac{1}{4}u(1+e)^2$, as required.

A third collision. C is faster than B (by $(*)$), so B and C separate. The only possible next collision is A catching B, which happens exactly when $v_A > v_B$:

$$v_A - v_B = \tfrac{1}{4}u(1-e)[2-(1+e)] = \tfrac{1}{4}u(1-e)^2 > 0,$$

since $e \neq 1$. So A is moving faster than B and behind it, and a third collision (A with B) occurs for every e in $0 < e < 1$.

Part (ii)

Third collision (A hits B): here $p = \frac{1}{2}u(1-e)$ and $q = \frac{1}{4}u(1-e^2)$, so

$$p+q = \tfrac{1}{4}u(1-e)[2+(1+e)] = \tfrac{1}{4}u(1-e)(3+e), \qquad p-q = \tfrac{1}{4}u(1-e)^2.$$

By $(*)$, the new velocity of B is

$$\begin{aligned} v_B &= \tfrac{1}{2}\left[\tfrac{1}{4}u(1-e)(3+e) + \tfrac{1}{4}eu(1-e)^2\right] \\ &= \tfrac{1}{8}u(1-e)[3+e+e-e^2] \\ &= \tfrac{1}{8}u(1-e)(3+2e-e^2) = \tfrac{1}{8}u(1-e)(1+e)(3-e), \end{aligned}$$

and (for completeness) $v_A = \frac{1}{8}u(1-e)[3+e-e(1-e)] = \frac{1}{8}u(1-e)(3+e^2)$, which is less than v_B.

Ask yourself After the third collision, which pairs could possibly collide again?

A and B have just collided, so by $(*)$ B is now faster than A and they are separating. C is still moving at $\frac{1}{4}u(1+e)^2$, and there is nothing beyond C. So a fourth collision happens if and only if B catches C, that is, if and only if $v_B > v_C$: dividing by $\frac{1}{8}u(1+e) > 0$,

$$\begin{aligned} \tfrac{1}{8}u(1-e)(1+e)(3-e) > \tfrac{1}{4}u(1+e)^2 &\iff (1-e)(3-e) > 2(1+e) \\ &\iff e^2 - 6e + 1 > 0. \end{aligned}$$

The quadratic $e^2 - 6e + 1$ has roots $e = 3 \pm 2\sqrt{2}$, and it is positive outside them. The larger root $3 + 2\sqrt{2}$ exceeds 1, while $3 - 2\sqrt{2} \approx 0.172$ lies in $(0, 1)$. So, for $0 < e < 1$,

$$\text{there is a fourth collision} \iff e < 3 - 2\sqrt{2}.$$

At $e = 3 - 2\sqrt{2}$ exactly, B and C move with equal velocity and never meet, so the inequality is strict.

Checks

- Momentum: after the second collision the total is $\frac{1}{4}u\left[2(1-e) + (1-e^2) + (1+e)^2\right] = \frac{1}{4}u \cdot 4 = u$, as it should be.
- Limiting cases: when $e \to 1$ the velocities after two collisions tend to 0, 0, u (the familiar "Newton's cradle" exchange) and $v_A - v_B \to 0$, so the third collision only just survives; the critical value $3 - 2\sqrt{2}$ lies well below 1, consistent with this.
- Numerically, with $e = 0.1$ the velocities after the third collision are $v_B \approx 0.3589u$ and $v_C = 0.3025u$, so B does catch C; with $e = 0.2$, $v_B = 0.336u < v_C = 0.36u$ and it does not. A computer simulation of the motion confirmed exactly three collisions for every $e > 0.1716$ and at least four below it.

Common slips

- Forgetting to justify *which* pair collides next. Each time, check that the two particles that have just collided are separating and that the remaining pair is closing.
- Getting the sign wrong in the restitution law: it is (speed of separation) $= e \times$ (speed of approach), so $q' - p' = e(p - q)$, with the front particle's velocity first on the left.
- Keeping the root $3 + 2\sqrt{2}$, which is outside the allowed range $0 < e < 1$.

Beyond STEP

The rule $(*)$ is a linear map taking the pair of velocities before a collision to the pair after, so a sequence of collisions is a product of matrices; asking how many collisions occur in total is a classic problem, with a beautiful geometric answer for elastic billiard-ball systems by "unfolding" the motion into a straight line in a wedge. For collisions, momentum and frames of reference treated with undergraduate care, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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